

- 4 (a) (i) amplitude = 0.2 mm A1 [1]
- (ii) period = 1.2 ms C1
frequency = 830 Hz A1 [2]
- (b) (i) any two of zero, 0.6 ms and 1.2 ms A1 [1]
- (ii) any two of 0.3 ms, 0.9 ms, 1.5 ms A1 [1]
- (c) *either* $v = \omega x_0 = 2\pi f x_0$
 $= 2\pi \times 830 \times 0.2 \times 10^{-3} = 1.05 \text{ m s}^{-1}$
or slope of graph = 1.0 m s^{-1} (allow $\pm 0.1 \text{ m s}^{-1}$) C1
 $E_K = \frac{1}{2}mv^2$
 $= \frac{1}{2} \times 2.5 \times 10^{-3} \times 1.05^2$ C1
 $= 1.4 \times 10^{-3} \text{ J}$ A1 [3]
- (d) (i) large / maximum amplitude of vibration B1
when impressed frequency equals natural frequency of vibration B1 [2]
- (ii) e.g. metal panels on machinery vibrate / oscillate (M1)
motor in machine impresses frequency on panel (A1)
e.g. car suspension system vibrates / oscillates (M1)
going over bumps would give large amplitude vibrations (A1)
any feasible example, M1 + A1 [2]

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5 (a) work done per / on unit positive charge